George Nigmatulin

PhD Researcher in Machine Learning and Quantitative Finance, University of Oxford

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Education

University of Oxford, DPhil in Engineering Science — Oct 2024 – Dec 2027 (exp.)
Machine Learning Research Group, Department of Engineering Science. Supervisors: Stefan Zohren and Jakob Foerster.
Thesis: "Generative AI for Financial Data: From Price-Impact Consistency to Reinforcement Learning Driven Trading".
ESRC Grand Union Doctoral Training Partnership (DTP) Data Skills Studentship.

Moscow State University, professional qualification: Financial Quantitative Analyst — Sep 2022 – Jan 2023

Moscow Institute of Physics and Technology, MSc in Applied Mathematics and Physics — Sep 2020 – Jun 2022
GPA 4.8/5.0, top 5% of the class. Thesis: "Anomaly Detection-Aided Active Learning on Smartwatch Data".

Orenburg State University, BSc in Applied Mathematics and Computer Science — Sep 2016 – Jun 2020
GPA 5.0/5.0, top 1% of the class. Thesis: "Machine Learning Models for Predicting and Optimizing ATM Service".

Experience

University of Oxford, Machine Learning Research Group · *Research Engineer* — Sep 2024 – present

- Built the LOB-Impact Benchmark, which tests whether generative limit order book models reproduce market impact: first-author paper at ICAIF 2026 (oral), open-source code.
- Study what state-space foundation models of the order book learn about the market (ICAIF 2026) and how generative models of order flow scale with data and compute.
- Co-developed JaxMARL-HFT, a GPU-accelerated multi-agent reinforcement learning environment for high-frequency trading with up to 240× faster training (ICAIF 2025, Honourable Mention).
- Member of BOLD, the British Open-ended Learning and Discovery lab; experiments on the Isambard-AI supercomputer.

Samsung R&D Institute · *Machine Learning Engineer, Sensor Solution Team* — Jul 2021 – Jun 2024

- Sweat loss estimation for smartwatches: accelerometer-based neural network for running distance (MAPE 7.7%, R^2 0.95) and kernel ridge regression for fluid loss (R^2 0.79). Published in IEEE Access.
- Regression models for body composition (body fat mass, skeletal muscle mass, intracellular and extracellular water) from a multi-frequency signal, released in Galaxy Watch 4.
- Body temperature algorithm: SVM heat detection on perfusion signals (accuracy 0.91, F1 0.87) and ridge regression for core temperature (R^2 0.77). US patent application.

Skolkovo Institute of Science and Technology · *Machine Learning Researcher* — Aug 2020 – Jul 2022

- Active learning guided by anomaly detection for experiment planning: 58% fewer labels and a 19% better regression model. Published at IEEE GLOBECOM.
- Uncertainty scores (total, data, knowledge) with Bayesian ensembles of decision trees and anomaly detection methods.

Bank Orenburg · *Data Scientist, IT Department* — Jun 2019 – Jul 2020

- Forecast daily ATM cash demand with exponential smoothing, ARIMA, neural networks and SSA; a dynamic-programming cash management model raised profitability by 30% over the classical model.

Publications

Nigmatulin, G., Mohl, V., Frey, S., Li, K., Marti-Shahandeh, A., Cucuringu, M., Calinescu, A., Foerster, J., Zohren, S. (2026). LOB-Impact Benchmark: Emergent Market Impact in Generative Limit Order Book Models. *ICAIF 2026*. **Oral.** arXiv:2610.27100

Mohl, V.*, Bismuth, A.*, Marti-Shahandeh, A., **Nigmatulin, G.**, Li, K., Zohren, S., Foerster, J., Cucuringu, M., Calinescu, A. (2026). Understanding the Inner Workings of a State-Space Foundation Model for Limit Order Books. *ICAIF 2026*. arXiv:2610.28753 (*equal contribution)

Nigmatulin, G., et al. (2026). Neural Scaling Laws for Order Flow Generation. Submitted to *NeurIPS 2026*.

Mohl, V., Frey, S., Leyland, R., Li, K., **Nigmatulin, G.**, Cucuringu, M., Zohren, S., Foerster, J., Calinescu, A. (2025). JaxMARL-HFT: GPU-Accelerated Large-Scale Multi-Agent Reinforcement Learning for High-Frequency Trading. *ICAIF 2025*, pp. 18–26. **Honourable Mention.** arXiv:2511.02136

Volkova, E., Pavlov, K., Beliaev, K., Perchik, A., Lychagov, V., Nikolaev, E., Ayuev, A., **Nigmatulin, G.**, Lee, W., Lee, H., Kim, Y. (2025). Electronic Device and Method for Determining a Body Core Temperature. US Patent Application US20250352075A1, Samsung Electronics.

Nigmatulin, G., Pavlov, K., Zaytsev, A. (2023). Anomaly Detection-Aided Active Learning. *IEEE GLOBECOM 2023*, pp. 1616–1619.

Pavlov, K., Perchik, A., Tsepulin, V., Megre, G., Nikolaev, E., Volkova, E., **Nigmatulin, G.**, Lee, H.J., Lee, W.S., Kim, Y.H. (2023). Sweat Loss Estimation Algorithm for Smartwatches. *IEEE Access*, 11, 23926–23934.

Nigmatulin, G., Chaganova, O. (2021). Research of an Optimization Model for Servicing a Network of ATMs and Information Payment Terminals. *Communications in Computer and Information Science*, Springer.

Bolodurina, I., **Nigmatulin, G.**, Parfenov, D. (2020). Intersection of Triangles in Space Based on Cutting Off Segment. *Communications in Computer and Information Science*, Springer.

Teaching

Deep Learning for Time Series Forecasting · Intelligent Earth CDT, University of Oxford Jan 2025, Jan 2026
Lab session in the Time Series Analysis and Signal Processing course: DeepAR and ConvTrans in PyTorch.

Volatility Forecasting with Economic Networks · The Oxford Summer School in Economic Networks Jun 2026
Tutorial with Yaxuan Kong: correlation networks of futures and a graph convolutional network for volatility.

Leadership and Service

BOLD Launch Festival, Rhodes House, Oxford Aug 2026

- Co-chair of the workshop *Collective Intelligence (incl. Prediction Markets) and the Optimal Co-Scientist*, with Yuhe Gao.
- Co-organiser of the workshop *Foundation Models and Open-Source in Finance* and moderator of its panel with Iain Dunning (Head of AI, Hudson River Trading), Stefan Zohren and Kang Li.

ICAIF 2025, Singapore: helped organise the Workshop on Rethinking Financial Time-Series Nov 2025

Reviewer: ICLR 2027, ICAIF 2026

Head of the Student Council, Orenburg State University 2018 – 2020

Skills

Python, JAX, PyTorch, C++ · English (C1), Russian (native)